

Correlations and Lags Graphs

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

corr_rows=273

lag_profiles_rows=1025

Interpretation option: inspect strongest co-movements and dominant lag distances.

Table: Corr_Matrix

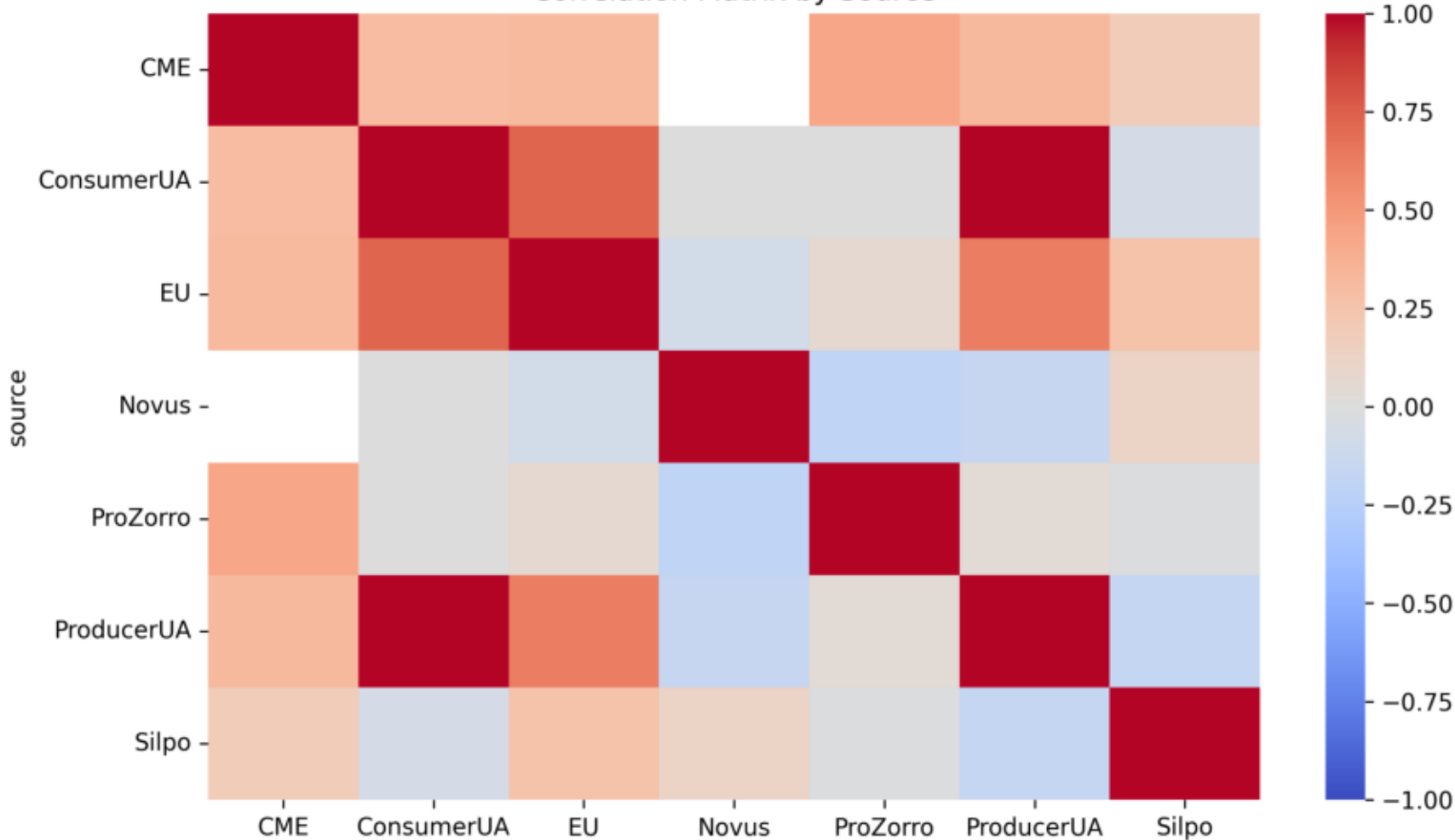
source	CME	ConsumerUA	EU	Novus	ProZorro	ProducerUA	Silpo
CME	1.0	0.30541603111957927	0.31350326247341676	nan	0.4309873151505198	0.3261217580643905	0.194237213355106
ConsumerUA	0.30541603111957927	1.0	0.7331215385198419	0.003320968190273582	0.006878099899064877	0.9931196793937417	-0.06557523486993877
EU	0.31350326247341676	0.7331215385198419	1.0	-0.0802936107605772	0.06313225435055486	0.6312783645531165	0.26221890542675513
Novus	nan	0.003320968190273582	-0.0802936107605772	1.0	-0.20114383934801688	-0.16236003168570914	0.11924189327160514
ProZorro	0.4309873151505198	0.006878099899064877	0.06313225435055486	-0.20114383934801688	1.0	0.03964845513664411	-0.009934571222856589
ProducerUA	0.3261217580643905	0.9931196793937417	0.6312783645531165	-0.16236003168570914	0.03964845513664411	1.0	-0.16734499420066853
Silpo	0.194237213355106	-0.06557523486993877	0.26221890542675513	0.11924189327160514	-0.009934571222856589	-0.16734499420066853	1.0

Table: Lag_Best

	product	standardized_type	pair_left	pair_right	lag_days	corr	is_best_lag
Масло	вершкове	butter	EU	ProZorro	28	-0.44070593855241375	1
Масло	вершкове	butter	EU	ProducerUA	1	0.7908068169800365	1
Масло	вершкове	butter	EU	Silpo	10	-0.25693434003769594	1
Масло	вершкове	butter	Novus	Silpo	24	-0.6048939452964793	1
Масло	вершкове	butter	ProducerUA	Novus	1	-0.1049819509606759	1
Масло	вершкове	butter	ProducerUA	ProZorro	20	-0.09608571873513129	1
Масло	вершкове	butter	ProducerUA	Silpo	8	-0.2508594893128858	1
Сир	кисломолочний	cottage_cheese	Novus	Silpo	10	0.44922039851458306	1
	Вершки	cream	EU	ProZorro	1	0.18843455826795666	1
	Вершки	cream	EU	ProducerUA	5	0.34445442438330837	1
	Вершки	cream	EU	Silpo	20	-0.4042612208416311	1
	Вершки	cream	Novus	Silpo	23	0.3696569431871244	1
	Вершки	cream	ProducerUA	Novus	10	0.19692193365974955	1
	Вершки	cream	ProducerUA	ProZorro	9	-0.08444260136425842	1
	Вершки	cream	ProducerUA	Silpo	15	-0.25412341759401047	1
Сир	твердий	hard_cheese	EU	ProZorro	11	-0.26333701523095143	1
Сир	твердий	hard_cheese	EU	ProducerUA	27	0.686007360451993	1
Сир	твердий	hard_cheese	EU	Silpo	24	-0.47036741987526826	1
Сир	твердий	hard_cheese	Novus	Silpo	7	-0.4139767649030091	1
Сир	твердий	hard_cheese	ProducerUA	ConsumerUA	1	0.9861527112372024	1
Сир	твердий	hard_cheese	ProducerUA	Novus	30	-0.35218643143168815	1
Сир	твердий	hard_cheese	ProducerUA	ProZorro	30	-0.112574663105113	1
Сир	твердий	hard_cheese	ProducerUA	Silpo	16	-0.42088622791084473	1
Інше/невідомо		milk	EU	Silpo	10	0.38129618404588617	1
	Кефір	milk	ProducerUA	Silpo	1	-0.4427213767975084	1
Молоко	питне	milk	EU	ProZorro	27	0.27851078423532943	1
Молоко	питне	milk	EU	ProducerUA	27	0.7065668481492933	1
Молоко	питне	milk	EU	Silpo	8	-0.39872005522637316	1
Молоко	питне	milk	Novus	Silpo	28	0.6180017302836935	1
Молоко	питне	milk	ProducerUA	ConsumerUA	1	0.9918066746547143	1
Молоко	питне	milk	ProducerUA	Novus	10	0.4093047790827283	1
Молоко	питне	milk	ProducerUA	ProZorro	19	-0.15415420640938843	1
Молоко	питне	milk	ProducerUA	Silpo	18	-0.3368524146103822	1
Інше/невідомо		other	EU	ProZorro	28	0.8726166637704476	1
Інше/невідомо		other	EU	Silpo	9	-0.5721977316688752	1
	Сметана	sour_cream	ProducerUA	ConsumerUA	1	0.9940901205039061	1
	Сметана	sour_cream	ProducerUA	ProZorro	14	0.09401258045745899	1
	Сметана	sour_cream	ProducerUA	Silpo	14	-0.29948795703794495	1
	Йогурт	yogurt_dessert	Novus	Silpo	21	-0.4606910370307454	1

corr_matrix_sources.png

Correlation Matrix by Source



lag_best_bar.png

Best Lag (days) by Product/Pair

